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Section: C

Differential Equation Assignment #3

Q) $f(x) = x$ $(0 < x < 2\pi)$

Let $x = \frac{1}{2} a_0 + a_1 \cos x + a_2 \cos 2x + b_1 \sin x +$

$b_2 \sin 2x + \dots \rightarrow \textcircled{1}$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} x dx$$

$$= \frac{1}{\pi} \left[\frac{x^2}{2} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[\frac{(2\pi)^2}{2} - \frac{0}{2} \right]$$

$$= \frac{1}{\pi} \left[\frac{4\pi^2}{2} \right]$$

$$\boxed{a_0 = 2\pi}$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos\left(\frac{n\pi x}{\pi}\right) dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} u \cdot v \cdot \cos nx \cdot dx$$

$$\int u v dx = uv_1 - u'v_2 \quad v_1 = \int v dx$$

$$a_n = \frac{1}{\pi} \left[x \left(\frac{2 \sin nx}{n} \right) - 1 \left(\frac{-\cos nx}{n^2} \right) \right]_0^{2\pi} \quad v_2 = \int v_1 dx$$

$$= \frac{1}{\pi} \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[\left(\frac{2\pi \sin n\pi}{n} + \frac{\cos 2n\pi}{n^2} \right) - \left(\frac{0 \sin 0}{n} + \frac{\cos(0)}{n^2} \right) \right]$$

$$= \frac{1}{\pi} \left[\frac{2\pi(0)}{n} + \frac{1}{n^2} - 0 - \frac{1}{n^2} \right]$$

$$= \frac{1}{\pi} [0 + 0 - 0]$$

$$\boxed{a_n = 0}$$

$$b_n =$$

$$\int u$$

$$b_n = \frac{1}{\pi}$$

$$= \frac{1}{\pi}$$

$$= \frac{1}{\pi}$$

$$= \frac{1}{\pi}$$

$$= \frac{1}{\pi}$$

$$b_n$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin\left(\frac{n\pi x}{\pi}\right) dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} \overset{u}{x} \overset{v}{\sin nx} dx$$

$$v_1 = \int v dx$$

$$v_2 = \int v_1 dx$$

$$\int u \cdot v dx = uv_1 - u'v_2$$

$$b_n = \frac{1}{\pi} \left[x \left(-\frac{\cos nx}{n} \right) - 1 \left(-\frac{\sin nx}{n^2} \right) \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[\frac{-x \cos nx}{n} + \frac{\sin nx}{n^2} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[\frac{-2\pi \cos 2n\pi}{n} + \frac{\sin 2n\pi}{n^2} \right] - \left(\frac{0 \cos(0)}{n} + \frac{\sin 0}{n^2} \right)$$

$$= \frac{1}{\pi} \left[\left(\frac{-2\pi(1)}{n} + \frac{0}{n^2} \right) - \left(\frac{0(1)}{n} + \frac{0}{n^2} \right) \right]$$

$$= \frac{1}{\pi} \left[-\frac{2\pi}{n} \right]$$

$$\boxed{b_n = -\frac{2}{n}}$$

$$\therefore n = 1, 2, 3, \dots$$

$$b_1 = -2, \quad b_2 = -1, \quad b_3 = -\frac{2}{3}, \dots$$

Now Substituting the values of a 's and b 's

$$x = \frac{1}{2} (2\pi) - 2 \sin x - \sin 2x - \frac{2}{3} \sin 3x - \dots$$

$$x = \pi - 2 \left(\sin x + \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x + \dots \right)$$

Ans

$$Q2) f(x) = -1 \quad (-1 < x < 0) \quad , \quad f(x) = 1 \quad (0 < x < 1)$$

$$P = 2L = 2$$

$$2L = 2$$

$$L = \frac{2}{2} = 1$$

$$\text{let } f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} \left\{ a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right\}, \dots$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$$

$$a_0 = \frac{1}{1} \int_{-1}^0 (-1) dx + \int_0^1 (1) dx$$

$$a_0 = [-x]_{-1}^0 + [x]_0^1$$

$$a_0 = [(0) - (-1)] + [(1) - (0)]$$

$$a_0 = 1 + 1$$

$$\boxed{a_0 = 2}$$

$$\sin n\pi = 0$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_n = \frac{1}{1} \int_{-1}^0 (-1) \cos(n\pi x) dx + \int_0^1 (1) \cos(n\pi x) dx$$

$$a_n = -1 \int_{-1}^0 \cos n\pi x dx + \int_0^1 \cos n\pi x dx$$

$D' = n\pi$ $D' = n\pi$

$$= \frac{-1}{n\pi} \int_{-1}^0 n\pi \cos n\pi x dx + \frac{1}{n\pi} \int_0^1 n\pi \cos n\pi x dx$$

$$= \frac{-1}{n\pi} \left[\frac{\sin n\pi x}{n\pi} \right]_{-1}^0 + \left[\frac{\sin n\pi x}{n\pi} \right]_0^1$$

$$= \frac{-1}{n\pi} \left[(\sin n\pi(0)) - (\sin n\pi(-1)) \right] + \frac{1}{n\pi} \left[(\sin n\pi(1)) - (\sin n\pi(0)) \right]$$

$$= \frac{-1}{n\pi} [0 - 0] + \frac{1}{n\pi} [0 - 0]$$

$$\boxed{a_n = 0}$$

$$\cos 0 = 1$$

$$\cos n\pi = (-1)^n$$

$$\cos -n\pi = (-1)^{-n}$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \left(\frac{n\pi x}{L} \right) dx$$

$$= \frac{1}{1} \int_{-1}^0 (-1) \sin(n\pi x) dx + \int_0^1 (1) \sin(n\pi x) dx$$

$$= -\frac{1}{1} \int_{-1}^0 \sin(n\pi x) dx + \int_0^1 \sin(n\pi x) dx$$

$$= -\frac{1}{n\pi} \int_{-1}^0 n\pi \cdot \sin(n\pi x) dx + \frac{1}{n\pi} \int_0^1 n\pi \cdot \sin(n\pi x) dx$$

$$= -\frac{1}{n\pi} \left[-\cos(n\pi x) \right]_{-1}^0 + \left[-\frac{\cos n\pi x}{n\pi} \right]_0^1$$

$$= \frac{1}{n\pi} \left[\cos(n\pi x) \right]_{-1}^0 + \frac{1}{n\pi} \left[\cos(n\pi x) \right]_0^1$$

$$= \frac{1}{n\pi} \left[(\cos(n\pi(0))) - (\cos(n\pi(-1))) \right] + \frac{1}{n\pi} \left[(\cos n\pi) - (\cos n\pi(0)) \right]$$

$$= \frac{1}{n\pi} \left[\cos(0) - (\cos(n\pi(-1))) \right] + \frac{1}{n\pi} \left[(-1)^n - \cos 0 \right]$$

$$= \frac{1}{n\pi} \left[1 - (-1)^{-n} \right] + \frac{1}{n\pi} \left[(-1)^n - 1 \right]$$

$$= \left[\frac{1 - (-1)^{-n}}{\pi n} \right] - \left[\frac{(-1)^n - 1}{\pi n} \right]$$

$$= \frac{(1 - (-1)^{-n}) - ((-1)^n - 1)}{\pi n}$$

$$= \frac{1 - (-1)^{-n} + 1 - (-1)^n}{\pi n}$$

$$b_n = \frac{2 - (-1)^{-n} - (-1)^n}{\pi n}$$

$$\therefore n = 1, 2, 3, \dots$$

$$n=1$$

$$b_1 = \frac{2 - (-1)^{-1} - (-1)^1}{\pi(1)}$$

$$b_1 = \frac{4}{\pi}$$

$$n=2$$

$$b_2 = \frac{2 - (-1)^{-2} - (-1)^2}{\pi(2)}$$

$$b_2 = 0$$

$$n=3$$

$$b_3 = \frac{2 - (-1)^{-3} - (-1)^3}{\pi(3)}$$

$$b_3 = \frac{4}{3\pi}$$

$$n = 4$$

$$b_4 = \frac{2 - (-1)^{-4} - (-1)^4}{\pi(4)}$$

$$b_4 = 0$$

$$n = 5$$

$$b_5 = \frac{2 - (-1)^{-5} - (-1)^5}{\pi(5)}$$

$$b_5 = \frac{4}{5\pi}$$

So,

$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} \left\{ a_n \cos \left(\frac{n\pi x}{L} \right) + b_n \sin \left(\frac{n\pi x}{L} \right) \right\}, \dots$$

$$= \frac{1}{2} (2) + \sum_{n=1}^{\infty} \left\{ (0) \cos \left(\frac{n\pi x}{L} \right) + b_n \sin \left(\frac{n\pi x}{L} \right) \right\}, \dots$$

$$= \frac{1}{\pi} + \frac{4}{\pi} \sin \left(\frac{(1)\pi x}{1} \right) + \frac{4}{3\pi} \sin \left(\frac{3\pi x}{1} \right) + \frac{4}{5\pi} \sin \left(\frac{5\pi x}{1} \right) + \dots$$

$$f(x) = \frac{1}{\pi} + \frac{4}{\pi} \left(\sin \pi x + \frac{1}{3} \sin 3\pi x + \frac{1}{5} \sin 5\pi x + \dots \right)$$

Ans

$$Q.3) f(x) = 1 - \frac{2}{L} x \quad (0 < x < L)$$

$$\text{let, } f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left\{ a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right\}$$

$$a_0 = \frac{1}{L} \int_0^L f(x) dx$$

$$a_0 =$$

$$\text{let, } f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}, \quad \text{---} \rightarrow \textcircled{1}$$

$$b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$$

$$b_n = \frac{2}{L} \int_0^L \left(1 - \frac{2}{L} x\right) \sin \frac{n\pi x}{L} dx \quad \text{---} \rightarrow \textcircled{2}$$

$$\int u \cdot v dx = uv_1 - u'v_2 \quad \text{---} \rightarrow \textcircled{3}$$

$$v_1 = \int v dx$$

$$= \int \frac{\sin n\pi x}{L} dx$$

$$D' = \frac{n\pi}{L}$$

$$= \frac{1}{n\pi/L} \int \frac{n\pi}{L} \sin \frac{n\pi}{L} x \, dx$$

$$v_1 = -\frac{L}{n\pi} \cos \frac{n\pi}{L} x$$

$$v_2 = \int v_1 \, dx$$

$$= \int -\frac{L}{n\pi} \cos \frac{n\pi}{L} x \, dx$$

$$= -\frac{L}{n\pi} \int \cos \frac{n\pi}{L} x \, dx$$

$$D' = \frac{n\pi}{L}$$

$$= -\frac{L}{n\pi} \times \frac{1}{\frac{n\pi}{L}} \int \frac{n\pi}{L} \cos \frac{n\pi}{L} x \, dx$$

$$v_2 = -\frac{L^2}{n^2 \pi^2} \sin \frac{n\pi}{L} x$$

$$u = \frac{1}{L} - \frac{2}{L} x$$

$$u' = 0 - \frac{2}{L}$$

putting v_1, v_2, u, u' in eq (3)

$$\int u \cdot v \, dx = uv, -u'v_2$$

$$\int u \cdot v \, dx = \left(\frac{1-2}{L} x \right) \left(\frac{-L}{n\pi} \frac{\cos n\pi x}{L} \right) - \left(\frac{-2}{L} \right) \left(\frac{-L^2}{n^2\pi^2} \frac{\sin n\pi x}{L} \right)$$

putting $\int u \cdot v \, dx$ in eq (2)

$$b_n = \frac{2}{L} \left[\left(\frac{1-2}{L} x \right) \left(\frac{-L}{n\pi} \frac{\cos n\pi x}{L} \right) - \left(\frac{-2}{L} \right) \left(\frac{-L^2}{n^2\pi^2} \frac{\sin n\pi x}{L} \right) \right]_0^L$$

$$b_n = \frac{2}{L} \left[\left(\left(\frac{1-2}{L} (L) \right) \left(\frac{-L}{n\pi} \frac{\cos n\pi (L)}{L} \right) - \left(\frac{-2}{L} \right) \left(\frac{-L^2}{n^2\pi^2} \frac{\sin n\pi (L)}{L} \right) \right) \right. \\ \left. - \left(\left(\frac{1-2}{L} (0) \right) \left(\frac{-L}{n\pi} \frac{\cos n\pi (0)}{L} \right) - \left(\frac{-2}{L} \right) \left(\frac{-L^2}{n^2\pi^2} \frac{\sin n\pi (0)}{L} \right) \right) \right]$$

$$b_n = \frac{2}{L} \left[\left(-1 \left(\frac{-L}{n\pi} \cos n\pi \right) - 0 \right) - \left(1 \left(\frac{-L}{n\pi} \cos 0 \right) - 0 \right) \right]$$

$$= \frac{2}{L} \frac{-L}{n\pi} [-\cos n\pi - \cos 0]$$

$$= -\frac{2}{n\pi} [-\cos n\pi - \cos 0]$$

$$b_n = \frac{-2}{n\pi} \left[-(-1)^n - 1 \right]$$

$$b_n = \frac{2}{n\pi} \left[(-1)^n + 1 \right]$$

$$\therefore n = 1, 2, 3, \dots$$

$$b_1 = \frac{2}{1\pi} \left[(-1)^1 + 1 \right]$$

$$b_1 = 0$$

$$b_2 = \frac{2}{2\pi} \left[(-1)^2 + 1 \right]$$

$$= \frac{2 \times 2}{2\pi}$$

$$b_2 = \frac{4}{2\pi}$$

$$b_3 = \frac{2}{3\pi} \left[(-1)^3 + 1 \right]$$

$$b_3 = 0$$

$$b_4 = \frac{2}{4\pi} \left[(-1)^4 + 1 \right]$$

$$b_4 = \frac{4}{4\pi}$$

$$b_5 = \frac{2}{5\pi} \left[(-1)^5 + 1 \right]$$

$$b_5 = 0$$

$$b_6 = \frac{2}{6\pi} \left[(-1)^6 + 1 \right]$$

$$b_6 = \frac{4}{6\pi}$$

Hence,

$$f(x) = \sum_{n=1}^{\infty} b_n \frac{\sin n\pi x}{L}$$

$$\frac{1-2x}{L} = \frac{4}{2\pi} \frac{\sin 2\pi x}{L} + \frac{4}{4\pi} \frac{\sin 4\pi x}{L} + \frac{4}{6\pi} \frac{\sin 6\pi x}{L}$$

$$\frac{1-2x}{L} = \frac{4}{\pi} \left(\frac{1}{2} \frac{\sin 2\pi x}{L} + \frac{1}{4} \frac{\sin 4\pi x}{L} + \frac{1}{6} \frac{\sin 6\pi x}{L} \right)$$

Ans

Q4) $f(x) = L - x \quad (0 < x < L)$

let $f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{L} x$ ——— (1)

$$b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi}{L} x \, dx$$

$$b_n = \frac{2}{L} \int_0^L (L-x) \left(\frac{\sin \frac{n\pi}{L} x}{L} \right) dx \quad \text{———— (2)}$$

$$\int u \cdot v \, dx = uv_1 - u'v_2 \quad \text{———— (3)}$$

$$v_1 = \int v \, dx$$

$$= \int \frac{\sin \frac{n\pi}{L} x}{L} \, dx$$

$$D' = \frac{n\pi}{L}$$

$$= \frac{1}{\frac{n\pi}{L}} \int \frac{n\pi}{L} \frac{\sin \frac{n\pi}{L} x}{L} \, dx$$

$$v_1 = \frac{L}{n\pi} \frac{\cos \frac{n\pi}{L} x}{L}$$

$$v_2 = \int v_1 dx$$

$$= \int \frac{-L}{n\pi} \frac{\cos n\pi x}{L} dx$$

$$D' = \frac{n\pi}{L}$$

$$= \frac{-L}{n\pi} \frac{1}{n\pi/L} \int \frac{n\pi}{L} \frac{\cos n\pi x}{L} dx$$

$$\boxed{v_2 = \frac{-L^2}{n^2\pi^2} \frac{\sin n\pi x}{L}}$$

$$u = (L-x)$$

$$u' = -1$$

putting v_1, v_2, u , and u' in eq (3)

$$\int u \cdot v dx = u v_1 - u' v_2$$

$$\int u \cdot v dx = (L-x) \left(\frac{-L}{n\pi} \frac{\cos n\pi x}{L} \right) - (-1) \left(\frac{-L^2}{n^2\pi^2} \frac{\sin n\pi x}{L} \right)$$

putting $\int u \cdot v dx$ in eq (2)

$$b_n = \frac{2}{L} \left[(L-x) \left(\frac{-L}{n\pi} \frac{\cos n\pi x}{L} \right) - (-1) \left(\frac{-L^2}{n^2\pi^2} \frac{\sin n\pi x}{L} \right) \right]_0^L$$

$$b_n = \frac{2}{L} \left[\left((L-L) \left(\frac{-L}{n\pi} \frac{\cos n\pi(L)}{L} \right) - (-1) \left(\frac{-L^2}{n^2\pi^2} \frac{\sin n\pi(L)}{L} \right) \right) - \left((L-0) \left(\frac{-L}{n\pi} \frac{\cos n\pi(0)}{L} \right) - (-1) \left(\frac{-L^2}{n^2\pi^2} \frac{\sin n\pi(0)}{L} \right) \right) \right]$$

$$= \frac{2}{L} \left[(0 + 0) - \left(L \left(\frac{-L}{n\pi} \cos 0 \right) + 0 \right) \right]$$

$$= \frac{2}{L} \left(\frac{L^2}{n\pi} \right)$$

$$\boxed{b_n = \frac{2L}{n\pi}}$$

$$\therefore n = 1, 2, 3, \dots$$

$$\boxed{b_1 = \frac{2L}{\pi}}$$

$$\boxed{b_2 = \frac{2L}{2\pi}}$$

$$\boxed{b_3 = \frac{2L}{3\pi}}$$

Hence,

$$f(x) = \sum_{n=1}^{\infty} b_n \frac{\sin n\pi x}{L}$$

$$L-x = \frac{2L}{\pi} \frac{\sin \pi x}{L} + \frac{2L}{2\pi} \frac{\sin 2\pi x}{L} + \frac{2L}{3\pi} \frac{\sin 3\pi x}{L} + \dots$$

$$= \frac{2}{\pi} L \left(\frac{\sin \pi x}{L} + \frac{1}{2} \frac{\sin 2\pi x}{L} + \frac{1}{3} \frac{\sin 3\pi x}{L} + \dots \right)$$

Ans